Lecture 1: AI/ML Foundations & Closed-Loop Control

Physics Foundations, Neural Surrogates, Reinforcement Learning, and Bayesian Optimization

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Why AI in Particle Accelerators?

Meeting stringent sub-micron stability tolerances while managing non-linear actuator dynamics

The Operational Challenge: High-Precision Orbit Stabilization

- **Extreme Beam Stability Tolerances:**

Modern synchrotron light sources, colliders, and free-electron lasers (FELs) require transverse orbit stabilization to within $< 10\%$ of beam size (sub-micron RMS tolerances).

- **High-Power Hadron Linacs (e.g., KOMAC, SNS, ESS):**

– Uncorrected trajectory offsets through RF accelerating cavities and focusing quadrupoles induce space-charge forces, emittance growth, and beam halo.
– Fractional beam losses exceeding 1 W/m ($\sim 10^{-5}$ of total beam power) cause severe structural radioactivation, prohibiting hands-on maintenance.

- **Limitations of Purely Classical Linear Control:**

– Analytical linear response matrices (ORM) ignore magnetic core saturation at high currents, inducing overshoot and limit cycles.
– Physical diagnostics carry 10 μm sensor noise floors; naive feedback amplifies measurement jitter into erratic actuator ripple.
– Multi-parameter coupling requires fast non-linear digital twins and adaptive closed-loop policies.

Program Schedule & Learning Objectives

1. Lecture 1: AI/ML Foundations & Closed-Loop Control (09:10 – 10:30)

- **Lecturer:** Chong Shik Park, Ph.D. (Korea University, Sejong)
- **Focus:** Transverse beam dynamics, Hill's equation, Orbit Response Matrix (ORM), non-linear cubic saturation, MLP surrogate modeling, SVD vs bounded inversion, RL with physics priors, and Bayesian optimization.

2. Lecture 2: Agentic AI for Accelerator Research (10:40 – 12:00)

- **Lecturer:** Thorsten Hellert, Ph.D. (Lawrence Berkeley National Laboratory / Osprey)
- **Focus:** LLM agents, dynamic Python API tool execution, automated scientific workflows, simulation orchestration, and human-in-the-loop operational safety guardrails.

3. Afternoon Mini-Hackathon: Beam Orbit Alignment Benchmark (13:30 – 17:00)

- 15-meter transport lattice, 12 BPMs, 6 steerers, 3 parallel challenge tracks (MLP, RL, BO).
- Real-time experiment tracking with Weights & Biases (W&B) and live global leaderboard.

AI Pillars: Surrogates & Optimization

Foundational applications revolutionizing beam physics simulations and online tuning

Pillars 1 & 2: Fast Digital Twins & Autonomous Multi-Objective Tuning

Pillar 1: Neural Surrogate Digital Twins (Fast Forward Simulation)

- High-fidelity particle tracking codes (`ELEGANT`, `MAD-X`, `OPAL`) simulate detailed collective space-charge forces, but require minutes to hours per run.
- Deep Neural Networks (MLPs, PINNs) learn compact, high-accuracy representations capable of sub-millisecond execution times ($<$ 0.2 ms).
- **Operational Impact:** Enables real-time model-predictive control (MPC), online orbit feedback, fast virtual diagnostics, and parameter sensitivity scans.

Pillar 2: Autonomous Multi-Objective Optimization & Tuning

- Accelerator operations require simultaneously balancing conflicting objectives: maximizing transmission and photon brightness while minimizing emittance and beam halo.
- Bayesian Optimization (BO) with Gaussian Processes leverages epistemic uncertainty and exploration-exploitation acquisition functions.
- **Operational Impact:** Achieves optimal machine operating points directly on physical beamlines in dozens of shots rather than thousands of manual scan steps.

AI Pillars: Diagnostics, Protection & Agents

Expanding AI applications into beam instrumentation, uptime preservation, and autonomous agents

Pillars 3, 4 & 5: Virtual Diagnostics, Protection & Agentic Control

Pillar 3: Virtual Diagnostics & Non-Destructive Phase Space Reconstruction

- Interceptive diagnostics (wire scanners, screens, deflecting cavities) destroy or disrupt the beam during measurement.
- Supervised ML reconstructs 6D phase space, bunch slice emittance, and current profiles shot-by-shot from non-invasive upstream RF probes and BPM readings.

Pillar 4: Anomaly Detection, Fault Prediction & Machine Protection

- Unscheduled downtime in multi-megawatt facilities incurs catastrophic financial and scientific operational costs.
- Unsupervised autoencoders detect precursor thermal drift; deep recurrent networks predict RF breakdown and superconducting cavity quenches milliseconds before failure.

Pillar 5: Agentic AI & Autonomous Operational Workflows

- Modern agent frameworks (e.g. *Osprey*) couple Large Language Model reasoning engines with standardized Python control APIs and simulation codes.
- Translates high-level natural language goals into validated, safe operational sequences under strict domain guardrails.

Derivation and Physical Interpretation of Hill's Equation

Governing Equation of Motion:

For an uncoupled linear magnetic transport line, particle displacement $x(s)$ along coordinate s satisfies:

$$\frac{d^2x(s)}{ds^2} + K(s)x(s) = 0$$

- **Quadrupole Focusing Function $K(s)$:**
  - $K(s) = \frac{e}{p}\frac{\partial B_y}{\partial x}$ defines the alternating gradient magnetic focusing strength.
  - $K > 0$ represents horizontally focusing quadrupoles (QF) that horizontally compress the beam envelope.
  - $K < 0$ represents horizontally defocusing quadrupoles (QD) that vertically focus the beam envelope.

- **Periodic Focusing Principle (Courant-Snyder Invariance):**

Even though $K(s)$ varies piecewise along the beam axis, alternating QF and QD pairs yield net focusing in both transverse planes simultaneously.

Betatron Motion & Courant-Snyder Optics

Analytical solution to Hill's equation and Courant-Snyder envelope parameters

Betatron Oscillation Solution & Envelope Dynamics

General Pseudo-Harmonic Betatron Solution:

The exact analytical solution to Hill's equation is expressed as a modulated harmonic wave:

$$x(s) = \sqrt{\epsilon \cdot \beta(s)} \cdot \cos\left(\psi(s) + \phi_0\right)$$

- **Physical Parameter Definitions:**
  - $\beta(s)$ [m]: Courant-Snyder beta function, describing the transverse beam envelope geometry along the lattice.
  - ϵ [m · rad]: Beam emittance, representing the conserved area occupied by the beam in transverse phase space (x, x').
  - $\psi(s) = \int_0^s \frac{d\bar{s}}{\beta(\bar{s})}$: The betatron phase advance accrued along the longitudinal coordinate.
  - $\nu = \frac{1}{2\pi} \oint \frac{ds}{\beta(s)}$: The betatron tune, representing the total number of transverse oscillations per machine turn.
- **Key Insight for Orbit Correction:**

Any localized angular deflection propagates downstream as a betatron oscillation modulated by $\sqrt{\beta(s)}$.

Orbit Response Matrix Derivation

Mathematical formulation linking localized steering kicks to downstream BPM readings

The Analytical ORM Transfer Function

Closed-Orbit Perturbation from a Localized Steerer Kick:

When a steering dipole at position s_j applies an angular kick $\theta_j = \frac{eBL}{p}$, the orbit shift at BPM position s_i is:

$$R_{ij} = \frac{\Delta x(s_i)}{\theta_j} = \frac{\sqrt{\beta_i \beta_j}}{2\sin(\pi\nu)} \cos\left(|\psi_i - \psi_j| - \pi\nu\right)$$

- **Global Matrix Superposition Formulation:**

For N corrector magnets and M Beam Position Monitors, the linear response is encapsulated as:

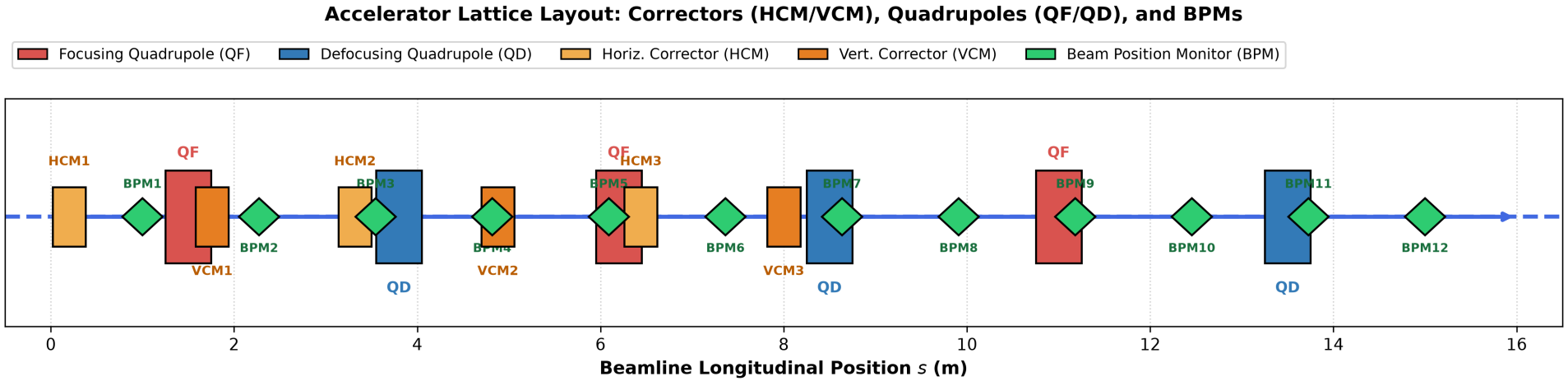
$$\mathbf{Y}_{\text{linear}} = \mathbf{I} \cdot \mathbf{R}$$

– $\mathbf{I} \in \mathbb{R}^6$: Vector of corrector coil currents in Amperes, mapped to kicks via magnetic field integration.
– $\mathbf{R} \in \mathbb{R}^{6\times12}$: The linear Orbit Response Matrix (ORM).
– $\mathbf{Y} \in \mathbb{R}^{12}$: Transverse displacements measured at all 12 downstream BPM stations.
– Cross-plane coupling: Any single corrector deflection alters the entire beam trajectory downstream.

HACKATHON BENCHMARK |

15-Meter Transport Lattice Architecture

Physical arrangement of quadrupole doublets, dipole correctors, and beam position monitors



Transport Line Specifications & Component Placement

- **12 Beam Position Monitors (BPMs):** Evenly distributed from $s = 1.0$ m to $s = 15.0$ m, measuring transverse displacement $\mathbf{Y}$ in millimeters.
- **6 Dipole Steering Magnets:** 3 Horizontal (HCM1-3) and 3 Vertical (VCM1-3) steerers positioned between $s = 0.2$ m and $s = 8.0$ m.
- **Alternating Quadrupole Doublets:** Pairs of QF and QD focusing magnets shaping the betatron phase advance and maintaining envelope stability.

Hardware Limits & Actuator Bounds

Realistic power supply operational boundaries and sensor measurement jitter

Power Supply Limits and Diagnostic Noise Specifications

1. Cumulative Corrector Current Bounds (Thermal & Magnetic Saturation Limit):

- Total current state: $I_k \in [-5.0\text{ A}, +5.0\text{ A}]$ for all 6 correctors.
- Magnets entering currents $|I| > 3.0$ A experience pronounced iron core saturation.
- Hardware protection interlocks trip power supplies if current demands exceed ± 5.0 A.

2. Per-Step Delta Current Bounds (Inductive Slew Rate Limit):

- Single-step control action: $\Delta I_k \in [-0.8\text{ A}, +0.8\text{ A}]$.
- Prevents high inductive back-EMF spikes ($L \cdot \frac{di}{dt}$) that stress magnet coils and trigger power supply voltage limits.

3. Gaussian Diagnostic Measurement Noise (Sensor Resolution Floor):

- Sensor noise: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, with $\sigma = 0.010$ mm (10 μm RMS resolution).
- **Crucial for control design:** Naive controllers misinterpret sensor jitter as physical orbit errors, commanding false kicks.

Mathematical Formulation of Non-Linear Beamline Response

Complete Beam Position Response Formulation:

The actual BPM observation vector $\mathbf{Y}$ in the simulation environment is governed by:

$$\mathbf{Y} = \mathbf{I} \cdot \mathbf{R} - 0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R}) + \mathbf{Y}_0 + \boldsymbol{\epsilon}$$

- **Physical Breakdown of Response Terms:**

– $\mathbf{I} \cdot \mathbf{R}$: Ideal linear betatron wave response predicted by transfer matrix.
– $-0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R})$: Non-linear iron core soft saturation. At high excitation currents, the iron yoke reaches magnetic saturation, decreasing steerer kick efficiency by 6% (at 2 A) up to 24% (at 4 A).
– $\mathbf{Y}_0$: Initial orbit distortion caused by random quadrupole transverse misalignments (producing 0.40 mm to 1.10 mm RMS error across seeds).
– $\boldsymbol{\epsilon}$: Diagnostic Gaussian measurement jitter ($\sigma = 10$ μm).

- **Convergence Target:** **Drive orbit error below RMS < 0.030 mm (30 μm) within 20 control steps.**

Neuron Transformation and Multi-Layer Composition

1. The Artificial Neuron Transformation:

Takes input vector $\mathbf{x} \in \mathbb{R}^d$, computes affine combination, and applies non-linear activation σ:

$$z = \sum_j w_j x_j + b = \mathbf{w}^T \mathbf{x} + b, \qquad a = \sigma(z)$$

2. Multi-Layer Perceptron (MLP) Deep Architecture:

Stacks L successive layers of affine transformations and non-linear activations:

$$\mathbf{h}^{(l)} = \sigma\left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right), \quad \textbf{with } \mathbf{h}^{(0)} = \mathbf{x}, \quad \hat{\mathbf{y}} = \mathbf{h}^{(L)}$$

3. Universal Approximation Theorem (Hornik et al., 1989):

• A feedforward network with at least one non-linear hidden layer can approximate any continuous non-linear function arbitrarily well, guaranteeing surrogate representability of magnetic saturation.

Activation Functions in Neural Networks

Comprehensive comparison of non-linear activations, gradients, and selection criteria

Core Activation Functions: Formula, Gradient & Use Cases

1. ReLU (Rectified Linear Unit): $f(z) = \max(0, z)$

- Constant gradient $= 1$ for $z > 0$. Prevents vanishing gradients, ultra-fast. Standard choice for surrogate hidden layers.

2. Leaky ReLU: $f(z) = \max(\alpha z, z), \quad \alpha \approx 0.01$

- Maintains a small non-zero slope for $z < 0$. Solves the "dying ReLU" issue where inactive neurons permanently shut down.

3. GELU (Gaussian Error Linear Unit): $f(z) = z \cdot \Phi(z)$

- Smooth probabilistic gating based on the standard normal CDF. State-of-the-art activation in modern LLMs & Transformers.

4. Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$

- Squeezes output to $(0, 1)$. Ideal for binary probability outputs; severe vanishing gradient when saturated at extremes.

5. Tanh (Hyperbolic Tangent): $\tanh(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$

- Zero-centered in $(-1, 1)$, offering stronger gradient dynamics than Sigmoid, but still susceptible to saturation at large $|z|$.

6. Linear / Identity: $f(z) = z$

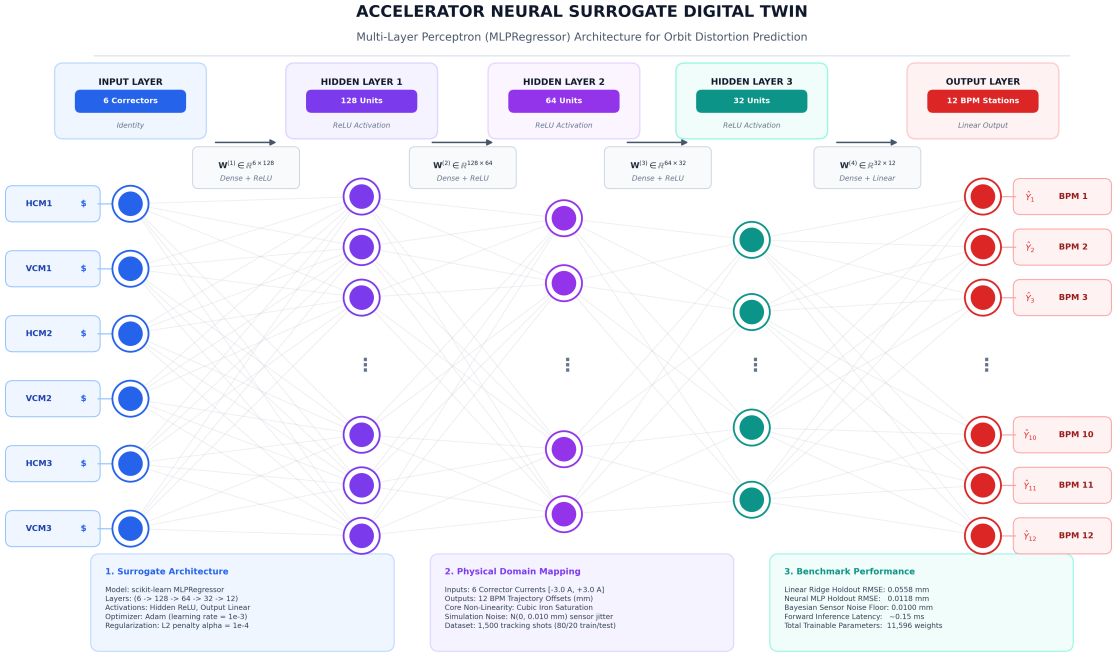
- No non-linearity. Essential for the output layer of continuous regression surrogates (e.g., predicting BPM readings $\mathbf{Y}$).

Evolution of Solvers: From Standard SGD to Adam and L-BFGS-B

1. **SGD (Stochastic Gradient Descent):** $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \cdot \mathbf{g}$
   - Basic first-order step. Prone to severe oscillation across ill-conditioned curvatures and high sensitivity to learning rate η.
2. **Momentum SGD:** $\mathbf{v} \leftarrow \gamma \mathbf{v} + \eta \mathbf{g}, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \mathbf{v}$
   - Accumulates velocity in directions of consistent gradients, dampening oscillations and accelerating through saddle points.
3. **AdaGrad (Adaptive Gradient):** $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\sqrt{\mathbf{G}+\epsilon}}\mathbf{g}$
   - Adapts learning rates per-parameter based on historical squared gradients. However, learning rate decays monotonically to zero.
4. **RMSProp:** $\mathbf{v} \leftarrow \beta \mathbf{v} + (1-\beta)\mathbf{g}^2, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\eta}{\sqrt{\mathbf{v}+\epsilon}}\mathbf{g}$
   - Uses exponentially decaying moving average of squared gradients. Solves AdaGrad's premature stopping problem.
5. **Adam (Adaptive Moment Estimation):** $\mathbf{m}_t$ **(1st moment)** $+ \mathbf{v}_t$ **(2nd moment)**
   - Combines Momentum and RMSProp with bias correction. Default gold standard for deep neural network surrogate training.
6. **AdamW (Decoupled Weight Decay):** **Direct** L_2 **weight decay subtraction**
   - Decouples L_2 penalty from adaptive gradient scaling. Dramatically improves generalization performance.
7. **L-BFGS-B (Quasi-Newton with Box Bounds):** **Second-order Hessian approximation**
   - Highly effective for bounded parameter inversion (Baseline 1 & Track 3) with tight box bounds $[-0.8, +0.8]$ A.

Neural Surrogate Digital Twin

Deep Multi-Layer Perceptron (128-64-32) capturing non-linear cubic saturation



- **Input Layer:** 6 Corrector Magnet Currents [HCM1-3, VCM1-3] in Amperes.
- **Hidden Layers:** Dense layers ($128 \rightarrow 64 \rightarrow 32$) with ReLU non-linear activations capturing cubic iron-core saturation.
- **Output Layer:** 12 predicted Beam Position Monitor readings [BPM1-12] in millimeters, evaluated in < 0.2 milliseconds.

Surrogate Training & Validation

Ensuring digital twin fidelity across linear and saturated magnetic operating regimes

Offline Dataset Generation and Model Verification

1. Synthetic Tracking Dataset Generation:

- 2,000 synthetic simulation runs generated by uniformly sampling corrector currents across $\mathbf{I} \in [-3.0, +3.0]^6$ A.
- Full non-linear response computed including the cubic soft-saturation term $-0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R})$.
- Dataset partitioned into 80% Training (1,600 samples) and 20% Holdout Testing (400 samples).

2. MLPRegressor Training Configuration:

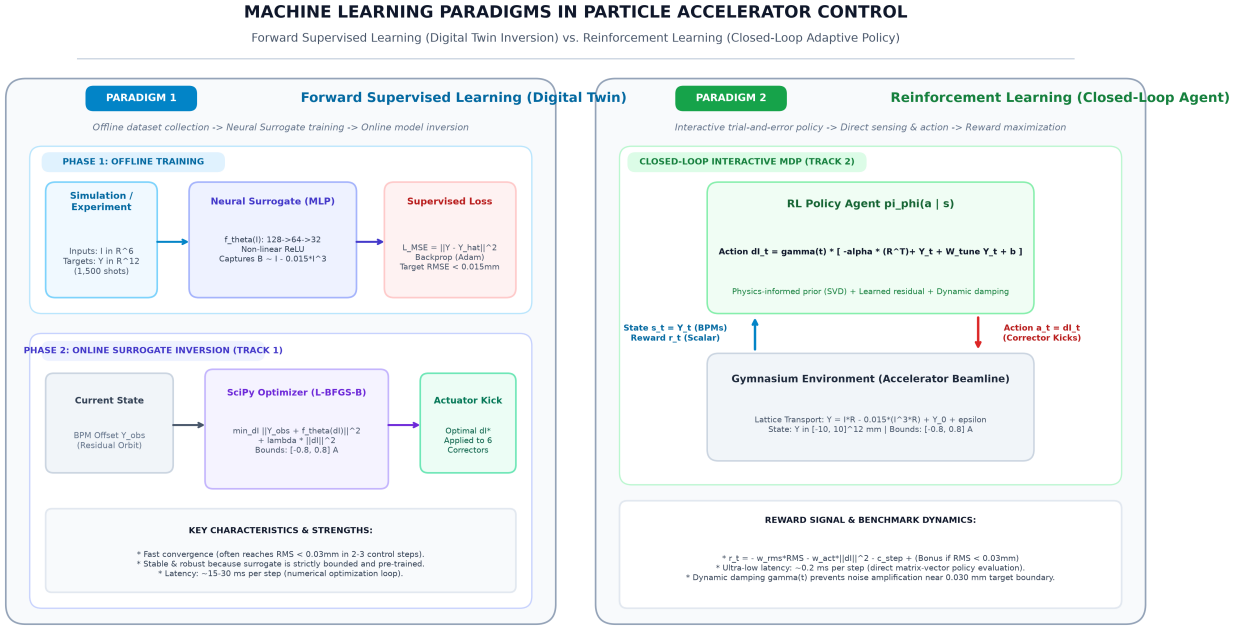
- Architecture: `hidden_layer_sizes=(128, 64, 32)`, `activation='relu'`, `solver='adam'`.
- Regularization: $\alpha = 10^{-4}$, `max_iter=500`, `early_stopping=True`.
- Loss convergence reached in under 180 epochs ($\sim$ 3.5 seconds CPU training time).

3. Holdout Verification Metrics:

- **Holdout Test MSE:** $< 2.5 \times 10^{-4}$ mm^2 across all 12 BPM stations.
- **Sub-Millisecond Execution:** Forward inference takes < 0.2 ms per call, 10,000$\times$ faster than numerical tracking.

Closed-Loop Control Architecture

Transitioning from predictive surrogate modeling to active closed-loop trajectory correction



- **Forward Supervised Model (Top):** Maps candidate corrector kick commands $\Delta\mathbf{I}$ to predicted BPM displacement vectors $\hat{\mathbf{Y}}$.
- **Closed-Loop Control Loop (Bottom):** Senses true BPM displacement $\mathbf{Y}$, passes error through controller policy, updates magnet power supplies, and repeats until convergence.

Baseline 0: Classical Linear SVD

Inverting the linear Orbit Response Matrix via Moore-Penrose pseudo-inverse

Mathematical Formulation & Operational Vulnerabilities

Singular Value Decomposition (SVD) Pseudo-Inverse Formulation:

Computes corrective kick vector $\Delta\mathbf{I}$ directly by inverting the transpose of the linear ORM:

$$\Delta\mathbf{I} = -\alpha \cdot (\mathbf{R}^T)^\dagger \cdot \mathbf{Y}_{\mathbf{obs}}, \qquad \textbf{with } (\mathbf{R}^T)^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T$$

- **Operational Profile & Limitations:**

– Static feedback gain: Typically $\alpha \approx 0.45$, with kicks hard clipped to $[-0.8, +0.8]$ A.
– Execution latency: < 0.1 ms (instantaneous analytical matrix-vector product).
– **Vulnerability in Non-Linear Regimes:** Completely oblivious to cubic iron core saturation. When magnets operate at high currents, the true response is much weaker than linear $\mathbf{R}$ predicts, leading to over-correction, slow convergence (5.6 mean steps), and high final current demands.

- **Tested Benchmark Performance: 100.0% success rate, 5.6 mean steps, 1,013.1 total score (Rank #3).**

Baseline 1: Supervised Neural Inversion

Model-based bounded optimization via SciPy L-BFGS-B on the trained neural digital twin

Optimization Formulation on Forward Surrogate

Per-Step Bounded Numerical Optimization Problem:

Instead of inverting an analytical matrix, Baseline 1 leverages the non-linear MLP twin $f_{\boldsymbol{\theta}}$:

$$\min_{\Delta \mathbf{I}} \mathcal{L}(\Delta \mathbf{I}) = \|\mathbf{Y}_{\mathbf{obs}} + f_{\boldsymbol{\theta}}(\Delta \mathbf{I})\|^2 + \lambda \|\Delta \mathbf{I}\|^2 \quad \textbf{subject to } \Delta \mathbf{I} \in [-5.0, +5.0]^6 \ \mathbf{A}$$

- **Why Supervised Inversion Outperforms Classical SVD:**
  - Because $f_{\boldsymbol{\theta}}$ captures the cubic saturation falloff, the optimizer automatically calculates higher current demands for saturated correctors.
  - Solver: SciPy `L-BFGS-B` converges in 8–15 iterations per control step ($\sim 15 - 25$ ms latency).
  - **Result:** Slashes convergence time from 5.6 steps (SVD) down to only 2.3 steps average!
- **Tested Benchmark Performance: 100.0% success rate, 2.3 mean steps, 1,016.8 total score (Rank #2).**

The Physics-Informed Residual Architecture

Hybrid Residual Policy Formulation:

Rather than learning control from scratch with unguided exploratory random kicks, our agent uses:

$$\Delta \mathbf{I}(t) = \gamma(t) \cdot \left[-\alpha (\mathbf{R}^T)^{\dagger} \mathbf{Y} + \mathbf{W}_{\mathbf{learned}} \mathbf{Y} + \mathbf{b} \right]$$

- **Architectural Synergy Breakdown:**

1. **Analytical Physics Prior ($-\alpha(\mathbf{R}^T)^{\dagger}\mathbf{Y}$):**
   – Anchors the policy in first-principles beam optics.
   – Guarantees immediate coarse stabilization from step 1 without needing millions of trial steps.
2. **Learned Residual Offset ($\mathbf{W}_{\mathbf{learned}}\mathbf{Y} + \mathbf{b}$):**
   – Neural network learns to predict non-linear saturation compensation.
   – Balances asymmetric horizontal (HCM) and vertical (VCM) steerer authorities dynamically.
3. **Execution Latency:** < 0.4 ms (pure feedforward matrix multiply, $50\times$ faster than iterative `L-BFGS-B`).

Dynamic Harmonic Step Damping

Preventing sensor-noise amplification and eliminating limit-cycle oscillations

Resolving the Sensor Noise Feedback Limit Cycle

The Noise Floor Bottleneck:

- As the orbit converges toward the sub-30 μm target, true orbit errors become comparable to BPM sensor noise ($\sigma = 10$ μm).
- Undamped linear feedback interprets stochastic sensor jitter as real trajectory error, commanding vigorous kicks that induce limit cycles around the boundary.

Harmonic Damping Schedule:

We modulate the policy kick authority with a dynamic temporal decay schedule:

$$\gamma(t) = \frac{1.0}{1.0+\kappa \cdot t}, \qquad \textbf{with } \kappa \approx 0.03$$

- **Step 1** ($t = 0, \gamma = 1.0$)**:** Full kick authority rapidly suppresses large millimeter-scale initial distortions.
- **Step 5+** ($t \geq 4, \gamma < 0.88$)**:** Attenuated feedback prevents sensor jitter amplification, locking the orbit cleanly below 0.030 mm.

Probabilistic Gaussian Process Modeling

Gaussian Process Surrogate Formulation:

Instead of deterministic point predictions, a GP models a distribution over candidate orbit response functions:

$$\hat{\mathbf{Y}}(\Delta\mathbf{I}) \sim \mathcal{GP}\left(\mathbf{m}(\Delta\mathbf{I}), \mathbf{K}(\Delta\mathbf{I}, \Delta\mathbf{I}')\right)$$

- **Composite Physics-Informed Kernel Architecture:**

$$k(\mathbf{x}, \mathbf{x}') = k_{\textbf{Matérn-5/2}}(\mathbf{x}, \mathbf{x}') + \sigma^2_{\textbf{noise}} \cdot \delta(\mathbf{x}, \mathbf{x}')$$

– **Matérn-5/2 Kernel:** Captures smooth, non-linear magnetic saturation falloff without imposing the unrealistic infinite differentiability of RBF kernels.
– **White Noise Kernel:** Explicitly accounts for the 10 μm BPM diagnostic jitter floor, avoiding kernel overfitting to noise.

- **Epistemic Uncertainty Estimation:**

For any candidate kick $\Delta\mathbf{I}$, the GP provides both posterior mean $\boldsymbol{\mu}_{\text{GP}}(\Delta\mathbf{I})$ and standard deviation $\boldsymbol{\sigma}_{\text{GP}}(\Delta\mathbf{I})$.

Lower Confidence Bound Acquisition

Balancing exploration of non-linear saturation against exploitation of predicted orbit corrections

The LCB Acquisition Strategy

Lower Confidence Bound (LCB) Mathematical Formulation:

At each step, the controller selects kicks by minimizing the LCB acquisition criterion:

$$\alpha_{\mathbf{LCB}}(\Delta\mathbf{I}) = \mathbf{RMS}\left(\mathbf{Y}_{\mathbf{obs}} + \boldsymbol{\mu}_{\mathbf{GP}}(\Delta\mathbf{I})\right) - \kappa \cdot \overline{\boldsymbol{\sigma}}_{\mathbf{GP}}(\Delta\mathbf{I}) + \lambda\|\Delta\mathbf{I}\|^2$$

- **Parameter Interpretations:**

– $\boldsymbol{\mu}_{\mathrm{GP}}(\Delta\mathbf{I})$: Posterior mean predicted orbit deviation across all 12 BPM stations.
– $\overline{\boldsymbol{\sigma}}_{\mathrm{GP}}(\Delta\mathbf{I})$: Posterior epistemic uncertainty (average standard deviation across BPMs).
– $\kappa \geq 0$: Exploration coefficient. When $\kappa > 0$, the agent explores uncharted high-current regimes; when $\kappa = 0$, it exploits the calibrated posterior mean.
– $\lambda\|\Delta\mathbf{I}\|^2$: Regularization penalty that minimizes actuator heating and avoids saturation.

- **Exceptional Convergence Speed:** **When calibrated, Track 3 reaches the convergence threshold in an average of only 1.5 steps!**

The Farama Gymnasium Environment Interface

Standard Gymnasium Method Signatures:

```
obs, info = env.reset(seed=seed)
obs, reward, terminated, truncated, info = env.step(action)
```

- **Observation Space:** `Box(-10.0, +10.0, shape=(12,), dtype=float32)` (12 transverse BPM readings in mm).
- **Action Space:** `Box(-0.8, +0.8, shape=(6,), dtype=float32)` (6 per-step delta kicks $\Delta\mathbf{I}$ in Amperes).
- **Step Reward Function:**

$$r_t = -\mathbf{RMS}_{t+1} - 0.01 \cdot \|\Delta\mathbf{I}\|_2 + (50.0 \textbf{ if } \mathbf{RMS}_{t+1} < 0.030 \textbf{ mm else } 0.0)$$

- **Termination Criteria:** `terminated = True` when RMS < 0.030 mm (success); `truncated = True` when step ≥ 20 (budget exhausted).

EVALUATION PROTOCOL | # Standardized 30-Seed Scoring Formula

Multi-objective evaluation metric balancing convergence reliability, speed, precision, and power

The Official Hackathon Scoring Formula

Standardized Evaluation Suite: `seeds = range(1, 31)` (30 evaluation seeds)

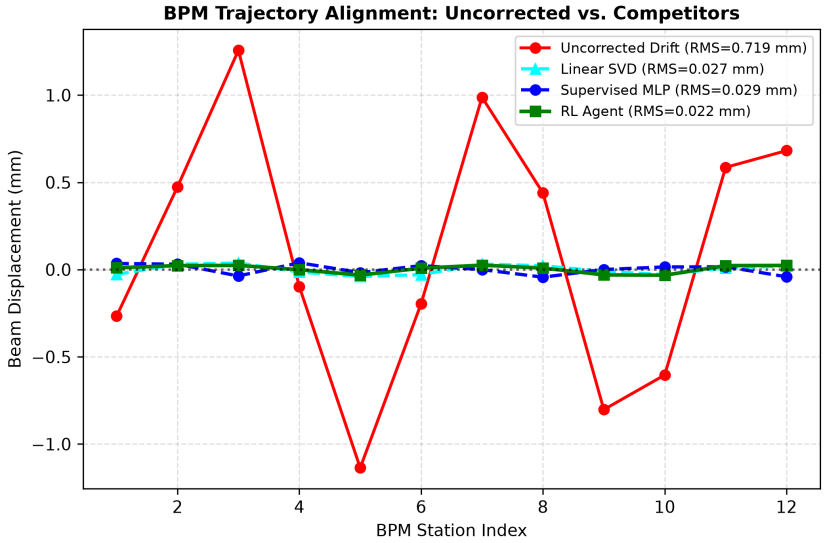
$$\textbf{Score} = (\textbf{Success Rate} \times 10.0) - (2.0 \times \overline{\textbf{Steps}}) + \left(\frac{0.03}{\max(\overline{\textbf{RMS}}, 0.005)} \times 20.0\right) - (2.0 \times \overline{\|\mathbf{I}\|})$$

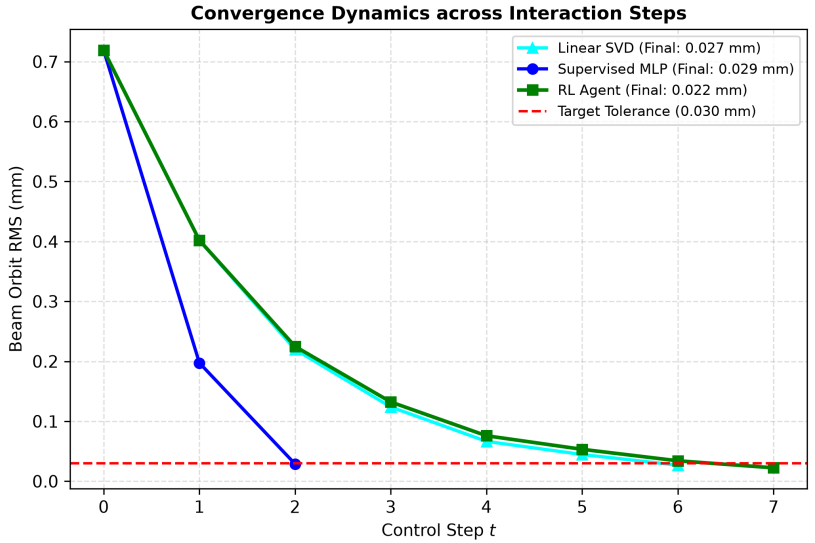
Breakdown of the Four Scoring Components:

1. **Success Rate** ($+10.0 \times \%$)**:** Rewards 100% convergence across all 30 seeds ($+1,000.0$ pts).
2. **Speed Penalty** ($-2.0 \times \overline{\textbf{Steps}}$)**:** Converging in 1.5 steps costs -3.0 pts, while 10 steps costs -20.0 pts.
3. **Sub-Micron Precision Bonus** ($+[\frac{0.03}{\textbf{RMS}}] \times 20.0$)**:** Achieving 0.015 mm yields $+40.0$ pts bonus.
4. **Current Economy Penalty** ($-2.0 \times \overline{\|\mathbf{I}\|}$)**:** Penalizes excessive steerer current norms to protect hardware.

Tested Benchmark Leaderboard & Trajectories

Verified comparative benchmark results across the 30 standardized evaluation seeds





Rank	Controller Paradigm	Success	Mean Steps	Final RMS	Current	Score
Rank 1	Track 3: Bayesian Optimization (BO)	**100.0%**	**1.5 steps**	**0.0149 mm**	**0.93 A**	**1,035.4 pts**
Rank 2	Baseline 1: Supervised Neural Inversion (MLP)	100.0%	2.3 steps	0.0257 mm	1.21 A	**1,016.8 pts**
Rank 3	Baseline 0: Classical Linear SVD	100.0%	5.6 steps	0.0230 mm	1.16 A	1,013.1 pts
Rank 4	Track 2: Closed-Loop RL Policy Agent	100.0%	6.1 steps	0.0239 mm	1.18 A	1,011.1 pts

Engineering Best Practices for Orbit Control

1. If Your Controller Oscillates or Diverges Near Target:

- Check your feedback gain α. Any gain > 0.6 will overcorrect when magnets enter saturated regions.
- Implement harmonic step damping: $\gamma(t) = 1.0/(1.0 + \kappa \cdot t)$. Attenuating authority at step 4+ eliminates sensor-noise limit cycles.

2. If Your Neural Surrogate Has High Generalization Error:

- Ensure your training dataset covers steerer excitations across $[-3.0, +3.0]$ A so that non-linear cubic saturation is well-sampled.
- Use ReLU activations and small L2 penalty ($\alpha = 10^{-4}$) to avoid overfitting.

3. Real-Time Telemetry Logging:

- Connect your notebook runs to Weights & Biases (`wandb.ai`) to visualize step-by-step RMS decay against reference baselines on the live leaderboard.

Key Takeaways & What Comes Next

Four Core Takeaways from Lecture 1:

1. **Physics Guides the AI:** Analytical optics priors (ORM pseudo-inverse) guarantee early stability and avoid millions of blind exploratory steps.
2. **Surrogates Capture Non-Linearities:** Compact deep MLPs accurately predict cubic core saturation, enabling 2-step bounded inversions.
3. **Probabilistic Uncertainty Wins:** Gaussian Process BO with LCB acquisition balances exploration against exploitation, leading tested scores in 1.5 steps.
4. **Dynamic Damping Prevents Limit Cycles:** Attenuating feedback near the 10 μm sensor noise floor locks residual orbit error.

Next Session (10:40 – 12:00):

Lecture 2: Agentic AI for Accelerator Research (Osprey Applications)

Speaker: Thorsten Hellert, Ph.D. (Lawrence Berkeley National Laboratory)
Topic: Autonomous operational agents, dynamic tool calling, and automated scientific workflows.

Q & A Discussion